PRODUCT PERFORMACE UPDATE

Application: SDI Pittsboro, Walking Beam Reheat Furnace

FURNACE DETAILS

- **Operation:** 125 TPH cold charge/165 TPH hot charge walking beam reheat furnace operating at 2350^{0}F, processing plain-carbon, low alloy and high-alloy billets & blooms for (SBQ) products.

- **Customer Need:** To minimize/reduce excessive Btu loss in skid pipes.

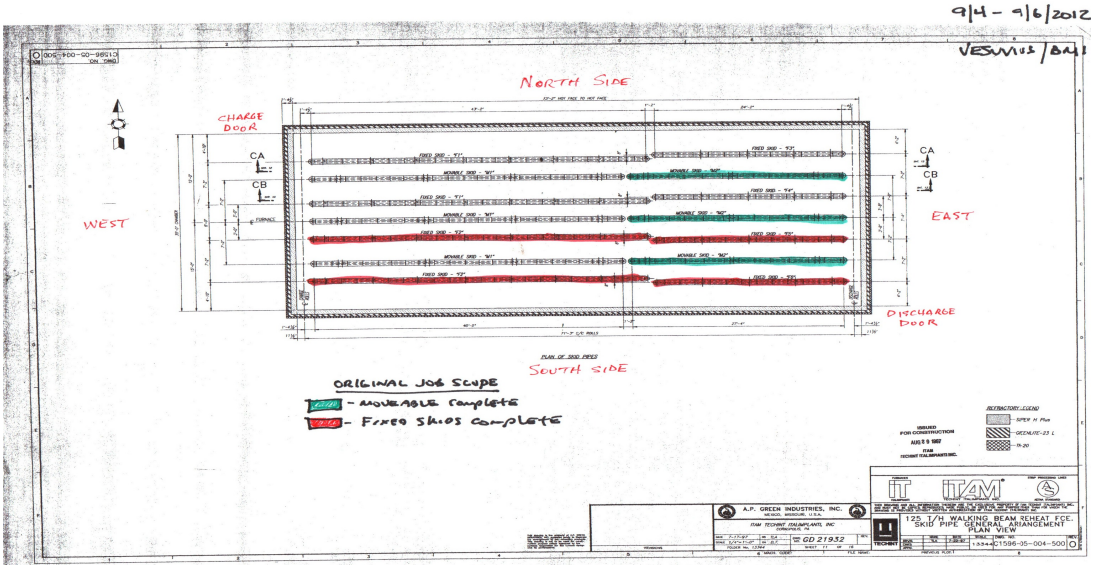
ITC RECOMMENDATION

- **Engineered Approach:** Using heat flow calculations, ITC recommended a hot face veneer to address Btu loss. The veneering design included;
  - 1" thick 12"x12" ceramic fiber squares on skid pipes
  - A proprietary, high temperature, energy efficient ceramic hot face coating

- **High Temperature Energy Efficient Ceramic Coatings:** ITC manufactures proprietary, high temperature energy efficient ceramic coatings to work in concert with the furnace conditions and veneer design. Under ideal conditions, high temperature, energy efficient ceramic coatings absorb energy from the process and re-radiate it to the furnace load, imparting the following benefits;

  Lower heat loss – Minimizes heat transfer through the furnace refractory walls
  Lower maintenance costs – Protects and prolongs the substrate refractory life
  Improved furnace operation – Improves the thermal efficiency of the process
  Reduction in scale generation – Less Fe_2O_3 available due to more efficient burning
  Increase thru put – Additional energy (Btu's) available to heat product at an increased rate

9/4 – 9/6/2012



IMPACT ON THE PROCESS

- LOWER DELTA T TEMEATURES
  The combination of veneer and the coating significantly lowered skid pipe delta T temperatures

- IMPROVED BTU SAVINGS
  Due to the thermal efficiencies, Btu savings were significantly better than the estimates

- REDUCED REFRACTORY MAINTENANCE
  The veneering system helped protect the skid pipe refractory shapes from thermal shock and heat related stresses therefore reducing overall maintenance required.

SDI - Pittsboro, Indiana

Bar Products Division

Vesuvius / BMI Reheat Furnace Veneering Project completed 9/4/2012 through 9/6/2012 - furnace started up on 9/7/2012 - temperature reading taken 9/13/12 by R Toth & R Darchuck

Chart Color	Location		Designation	Inlet Temperature	Outlet Temperature	Delta T	2011 Delta T	COMMENTS
	North Side of Furnace Skid & Posts - Charge/Heat Zone	Veneer only on missing areas	Moveable 1	93	100	7	26	
	Center of Furnace Skid & Posts - Charge/Heat Zone	Veneer only on missing areas	Moveable 1	92	100	8	35	
	South Side of Furnace Skid & Posts - Charge/Heat Zone	Veneer only on missing areas	Moveable 1	92	104	12	38	
	North Side of Furnace Skid & Posts - Soak Zone	Veneered full Skid & Posts	Moveable 2	93	101	8	20	Hottest zone of furnace
	Center of Furnace Skid & Posts- Soak Zone	Veneered full Skid & Posts	Moveable 2	92	103	11	14	Hottest zone of furnace
	South Side of Furnace Skid & Posts- Soak Zone	Veneered full Skid & Posts	Moveable 2	92	105	13	36	Hottest zone of furnace
	North Side of Furnace 2 -Fixed Skids - Charge/Heat/Soak Zone	Veneer only on missing areas	F1/F3/F4/F1	92	102	10	35	
	South Side of Furnace 2-Fixed Skids - Charge/heat/Soak Zone	Veneered full Skids	F2/F5/F6/F2	92	95	3	31	
	North Side of Furnace - Fixed Skids Posts - all zones	Veneer only on missing areas	F1 / F3	92				No reading taken
	Center North of Furnace - Fixed Skid Posts - all zones	Veneer only on missing areas	F1 / F4	92				No reading taken
	Center South of Furnace - Fixed Skid Posts - all zones	Veneered full Posts	F2 / F5	92				No reading taken
	South Side of Furnace - Fixed Skid Posts - all zones	Veneered full Posts	F2 / F6	92				No reading taken

*** Furnace set point at 2300F (2380F is max soak zone temp)
*** Furnace billet draw rate at average of 200 seconds per billet (170 is max)
*** Furnace running multiple grades of steel at different bar lengths

IMPACT ON BOTTOM LINE COSTS

- **Btu savings per Year**

  Yearly savings in natural gas associated with one (1) horizontal movable skid pipe,

TOTAL IMPACT/YEAR = $33,857.00